Dataset Validation

Surgical Scene Semantic Segmentation

Dataset: dsad

Number of classes: 8

Complete-dataset distribution analysis

This report evaluates class presence, pixel-level distribution, split consistency, and qualitative segmentation annotations.

All quantitative statistics are computed using the complete samples available in each split.

1. Dataset Overview

Dataset: dsad

Split	Number of samples
train	863
validation	202
test	365

Classes

```
0  —  background
1  —  abdominal_wall
2  —  colon
3  —  liver
4  —  pancreas
5  —  small_intestine
6  —  spleen
7  —  stomach
```

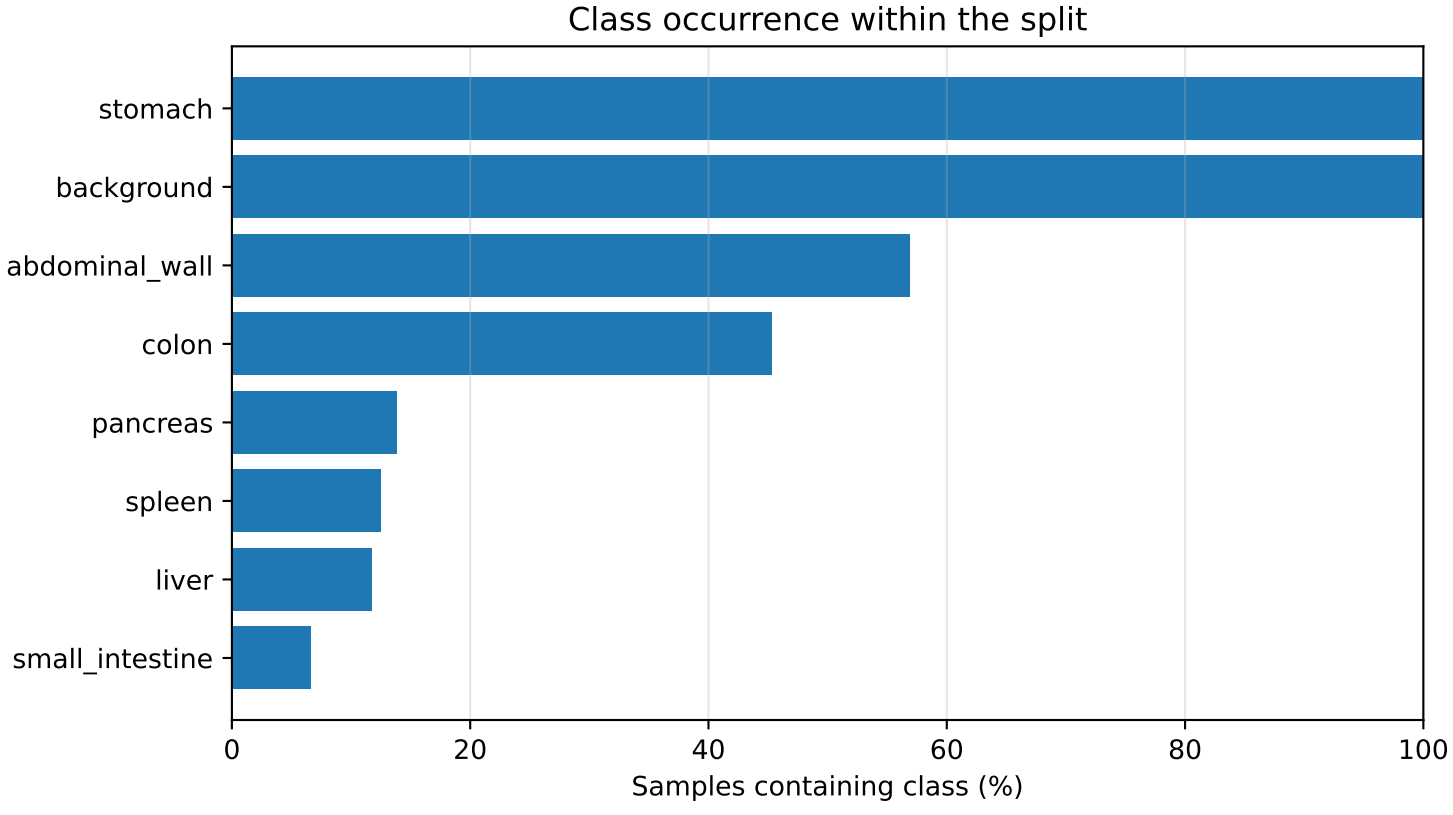
Methodology

The validation procedure evaluates every sample in each split. No sampling or maximum-sample limit is applied to the quantitative analysis.

2. Class Presence

Split: train

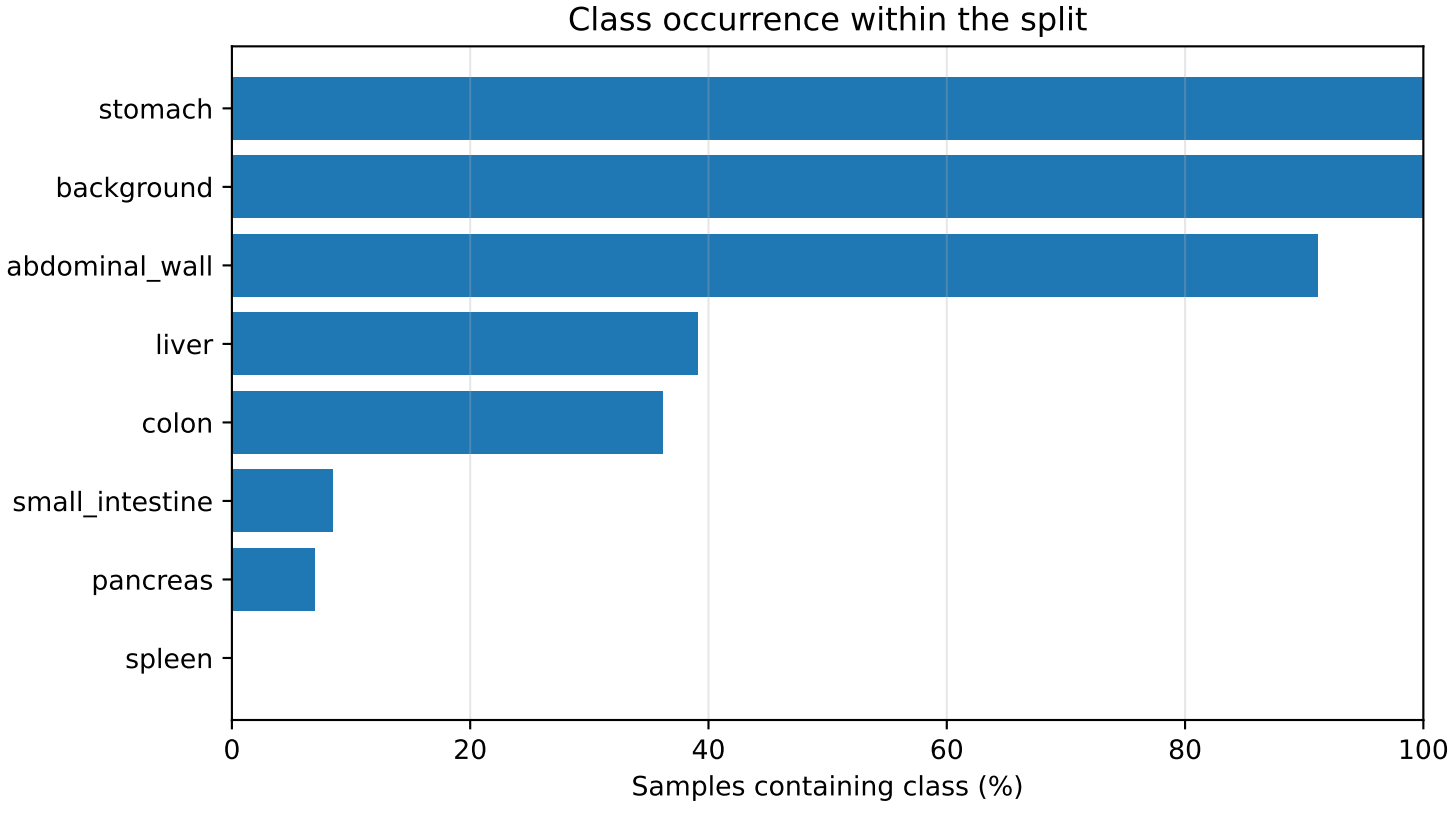
Class	Samples	Samples containing class
background	863	100.00%
abdominal_wall	491	56.89%
colon	391	45.31%
liver	101	11.70%
pancreas	119	13.79%
small_intestine	57	6.60%
spleen	108	12.51%
stomach	863	100.00%



2. Class Presence

Split: validation

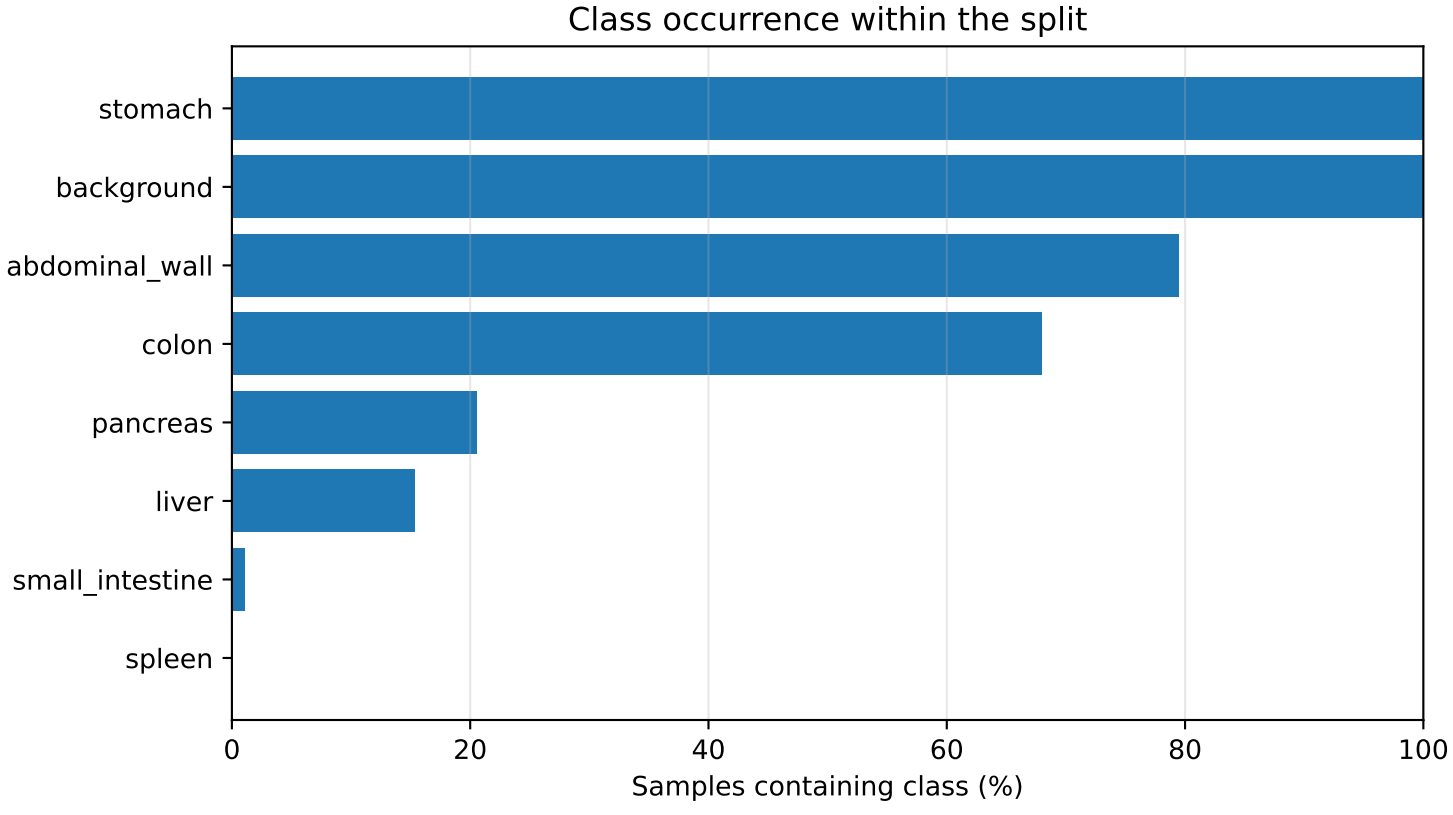
Class	Samples	Samples containing class
background	202	100.00%
abdominal_wall	184	91.09%
colon	73	36.14%
liver	79	39.11%
pancreas	14	6.93%
small_intestine	17	8.42%
spleen	0	0.00%
stomach	202	100.00%



2. Class Presence

Split: test

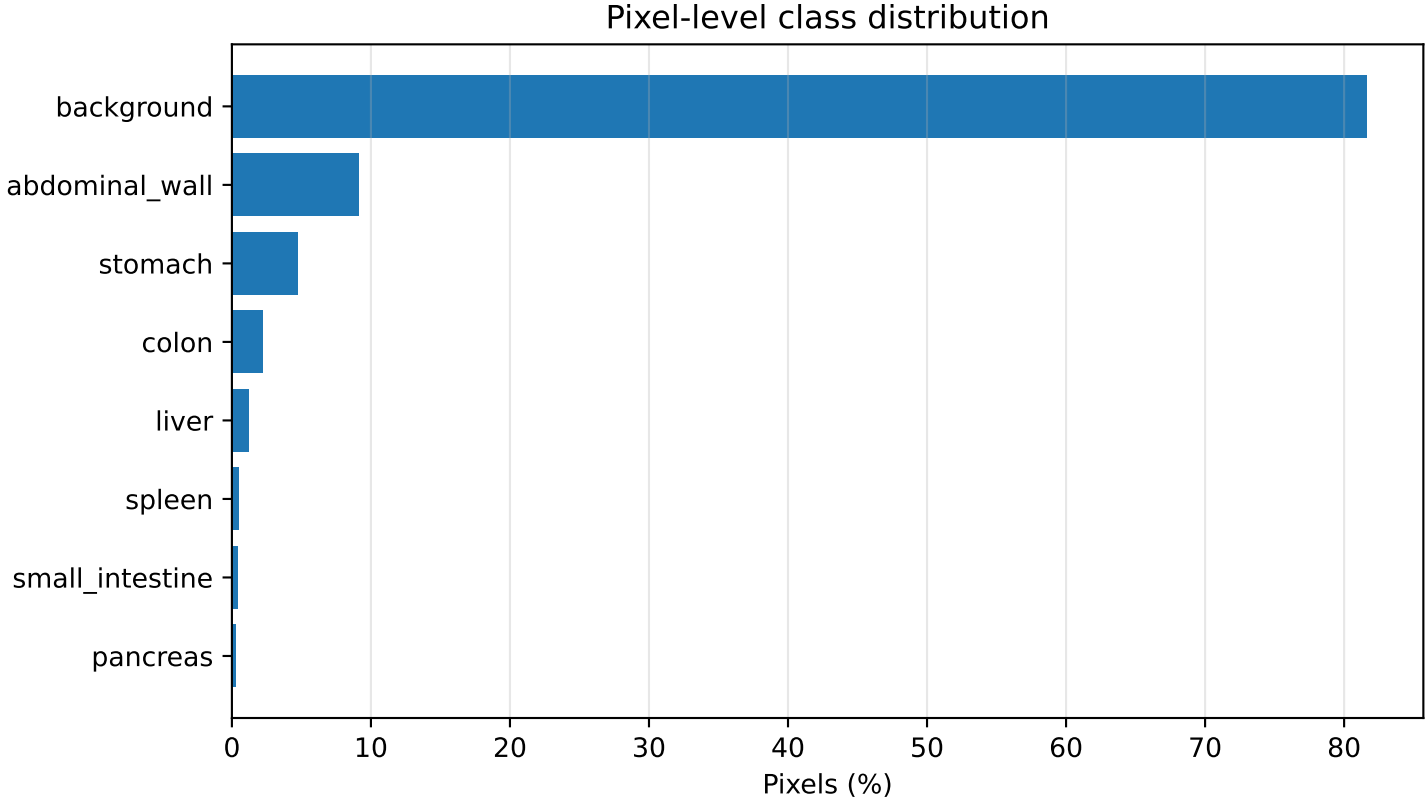
Class	Samples	Samples containing class
background	365	100.00%
abdominal_wall	290	79.45%
colon	248	67.95%
liver	56	15.34%
pancreas	75	20.55%
small_intestine	4	1.10%
spleen	0	0.00%
stomach	365	100.00%



3. Pixel Distribution

Split: train

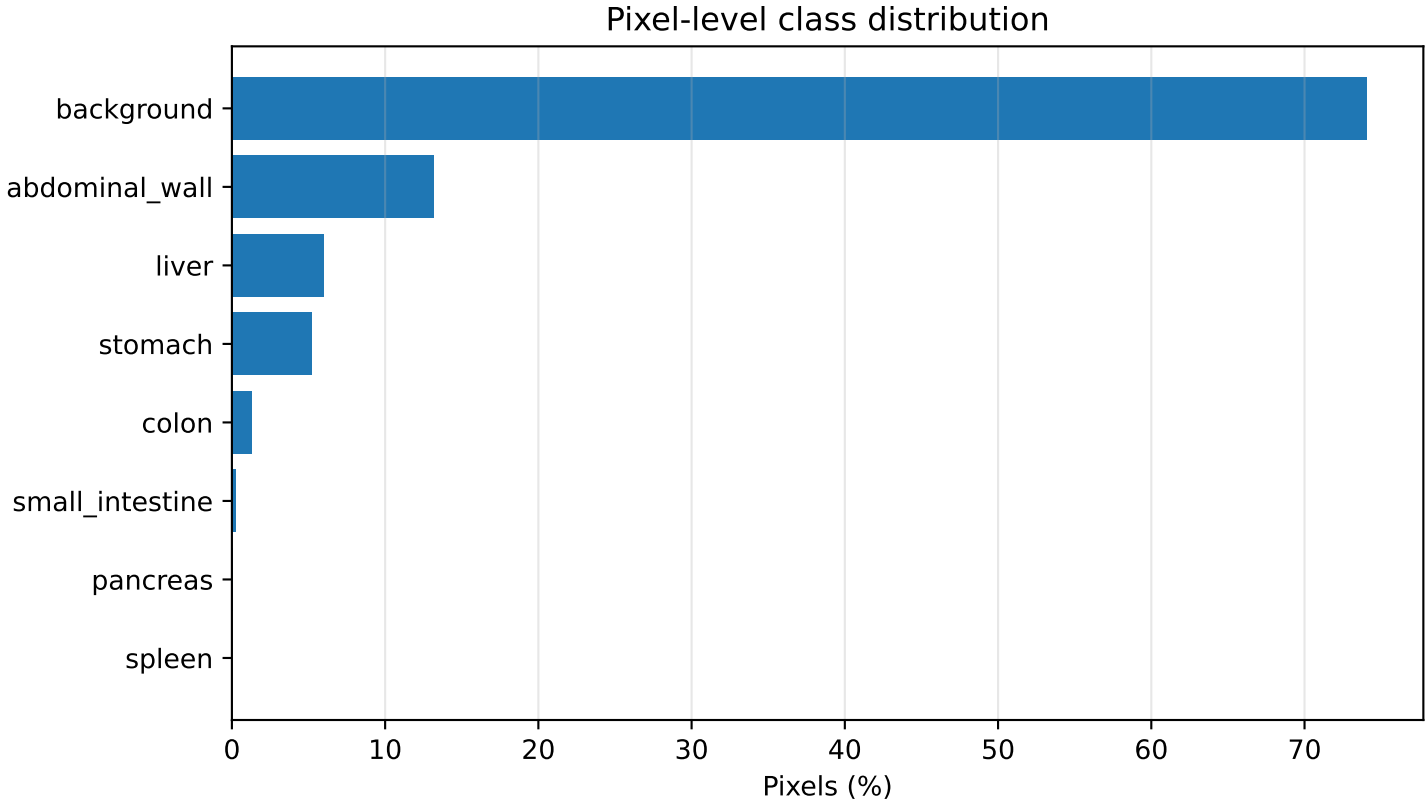
Class	Pixels	Pixel distribution
background	923,230,982	81.6187%
abdominal_wall	102,713,874	9.0805%
colon	24,907,744	2.2020%
liver	13,796,672	1.2197%
pancreas	2,798,809	0.2474%
small_intestine	4,609,369	0.4075%
spleen	5,659,162	0.5003%
stomach	53,434,748	4.7239%



3. Pixel Distribution

Split: validation

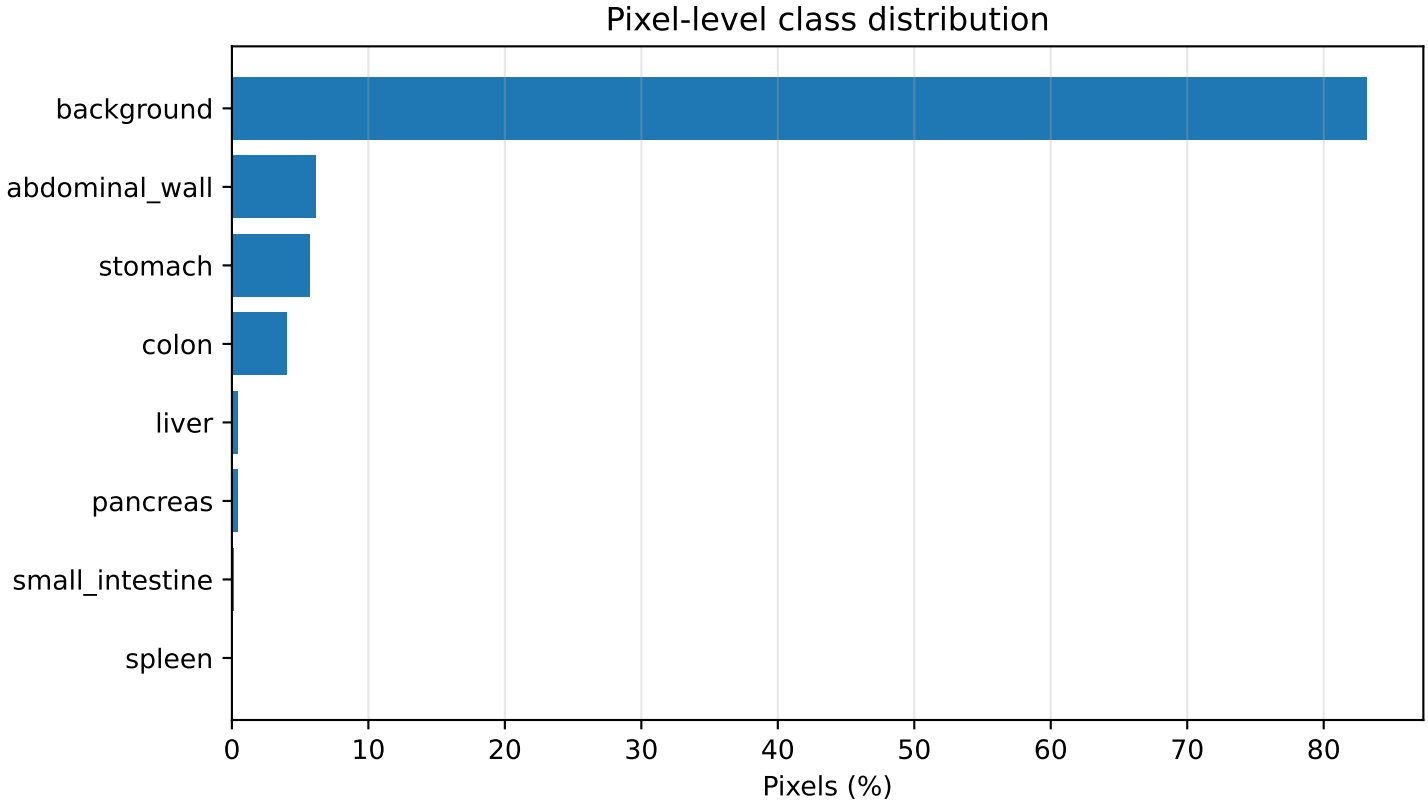
Class	Pixels	Pixel distribution
background	196,063,774	74.0519%
abdominal_wall	34,864,733	13.1682%
colon	3,481,432	1.3149%
liver	15,767,107	5.9551%
pancreas	99,303	0.0375%
small_intestine	714,571	0.2699%
spleen	0	0.0000%
stomach	13,774,520	5.2025%

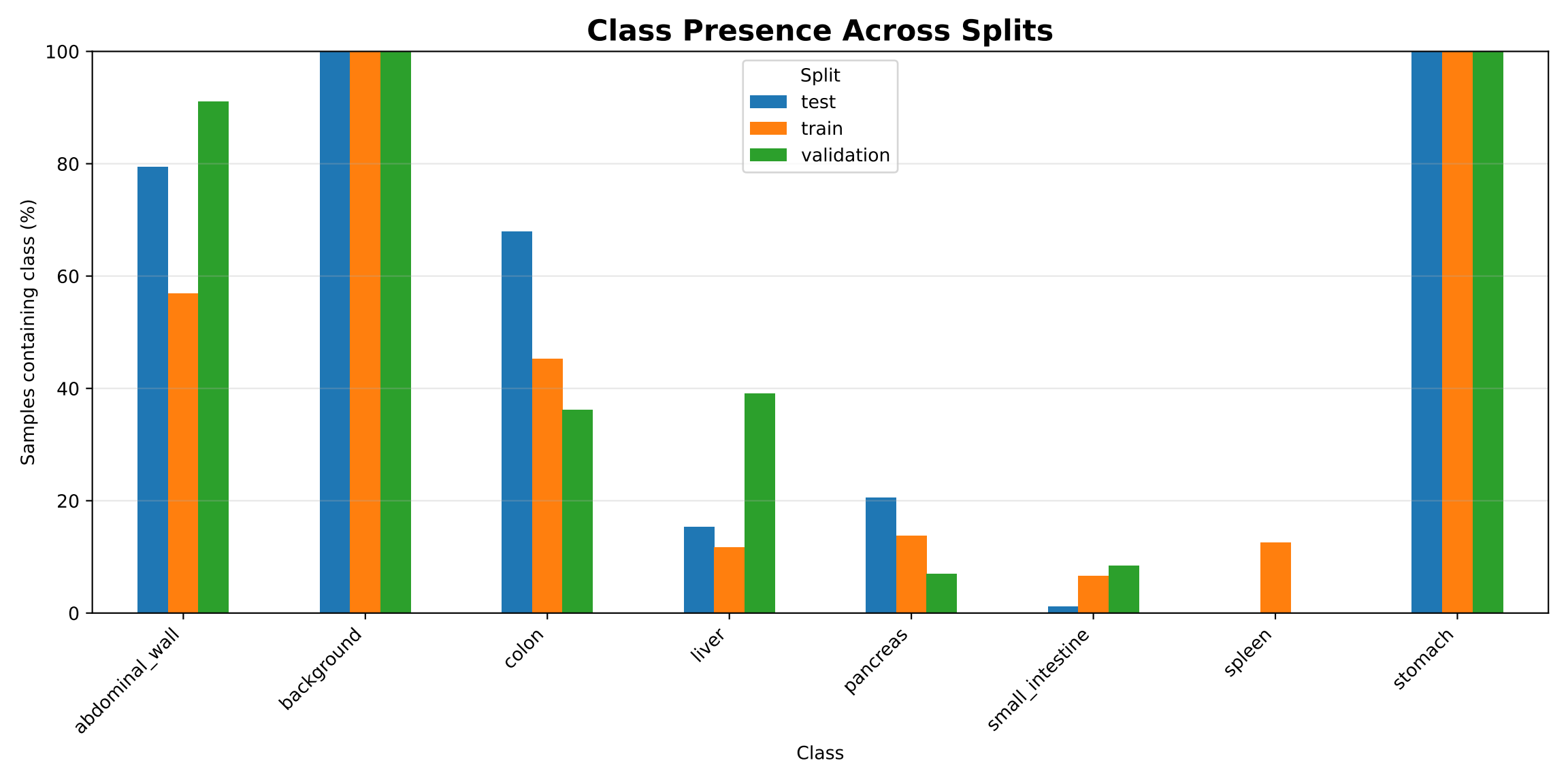


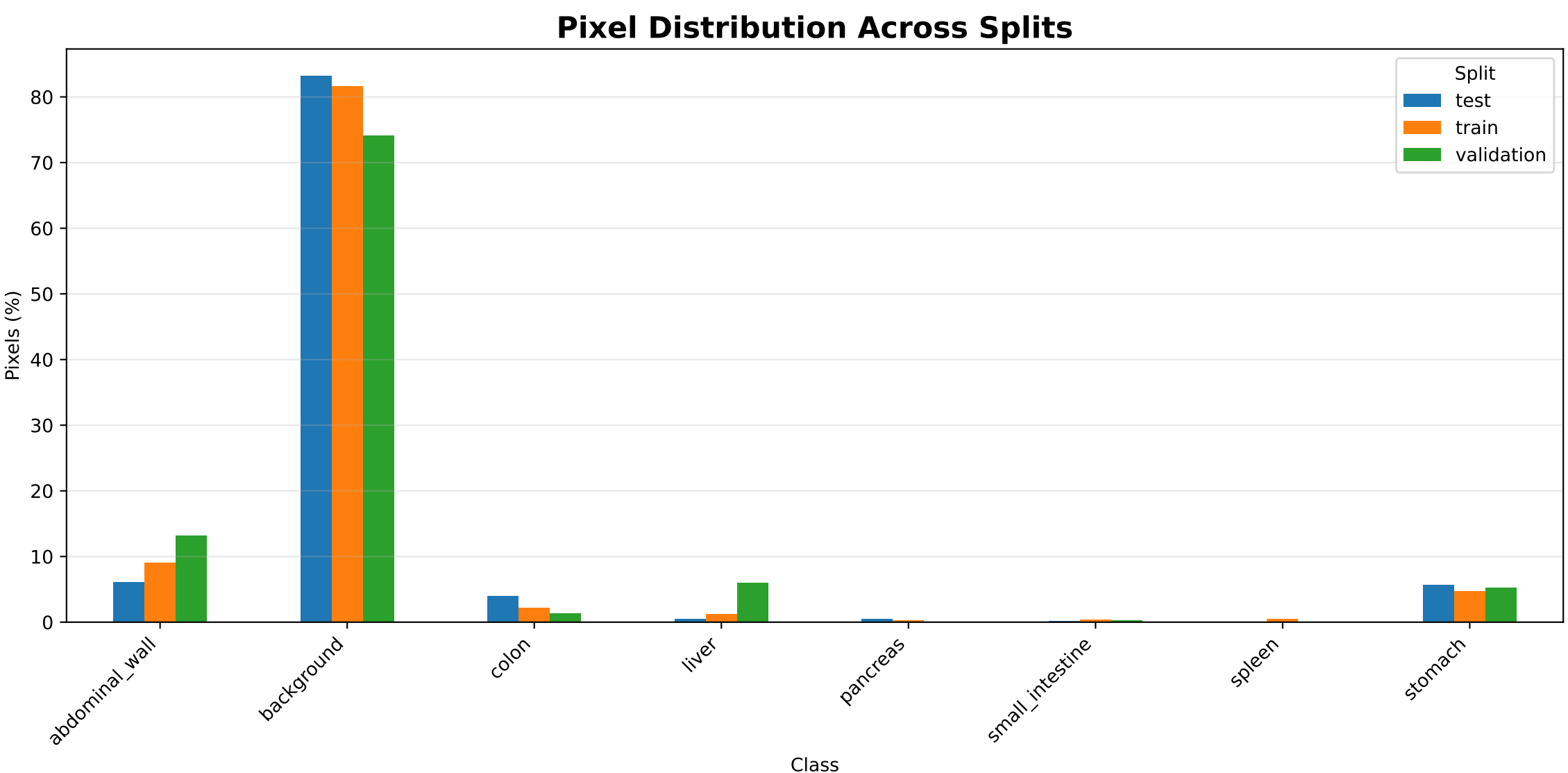
3. Pixel Distribution

Split: test

Class	Pixels	Pixel distribution
background	397,771,666	83.1440%
abdominal_wall	29,272,505	6.1187%
colon	19,217,338	4.0169%
liver	2,141,406	0.4476%
pancreas	2,085,896	0.4360%
small_intestine	732,359	0.1531%
spleen	0	0.0000%
stomach	27,191,630	5.6837%

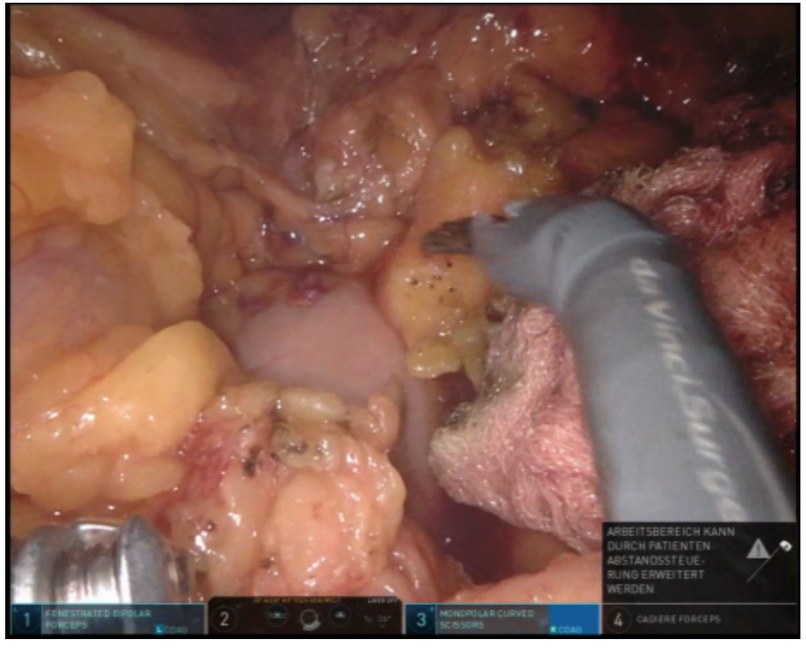



Class Presence Across Splits
Split
test
train
validation
Samples containing class (%)
0
20
40
60
80
100
abdominal_wall
background
colon
liver
pancreas
small_intestine
spleen
stomach
Class


Pixel Distribution Across Splits
Split
test
train
validation
Pixels (%)
0
10
20
30
40
50
60
70
80
abdominal_wall
background
colon
liver
pancreas
small_intestine
spleen
stomach
Class

4. Qualitative Analysis — train — Sample 665

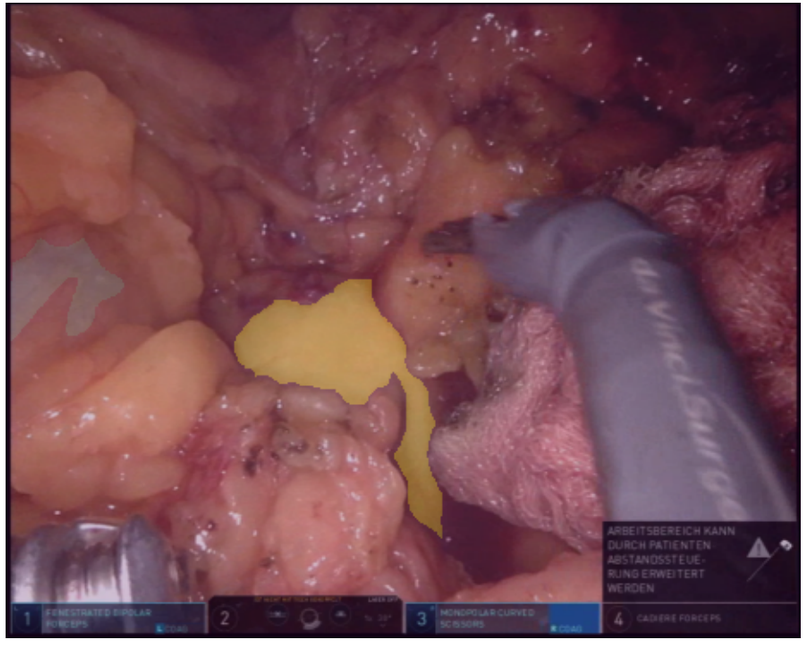
Reference Image



Ground Truth



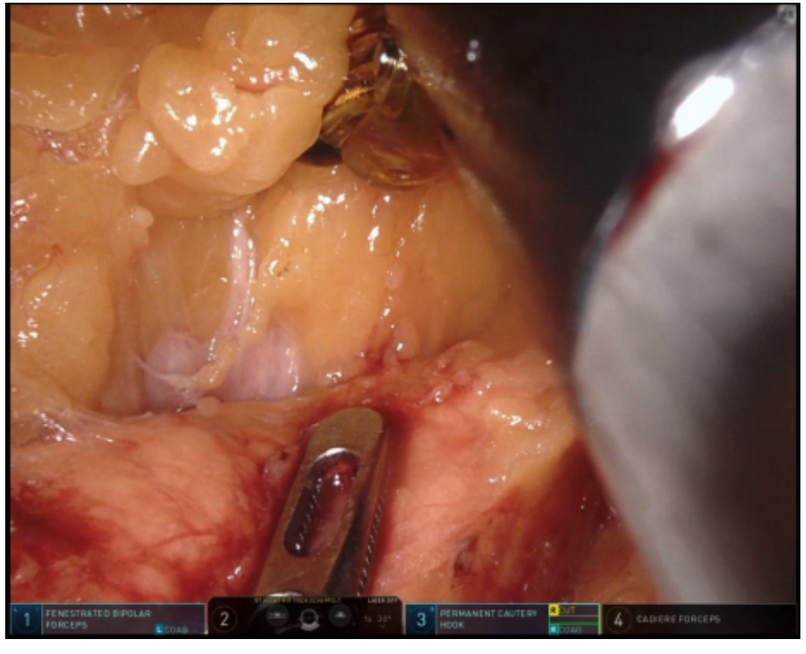
Ground Truth Overlay



Present classes: background, colon, stomach

4. Qualitative Analysis — train — Sample 378

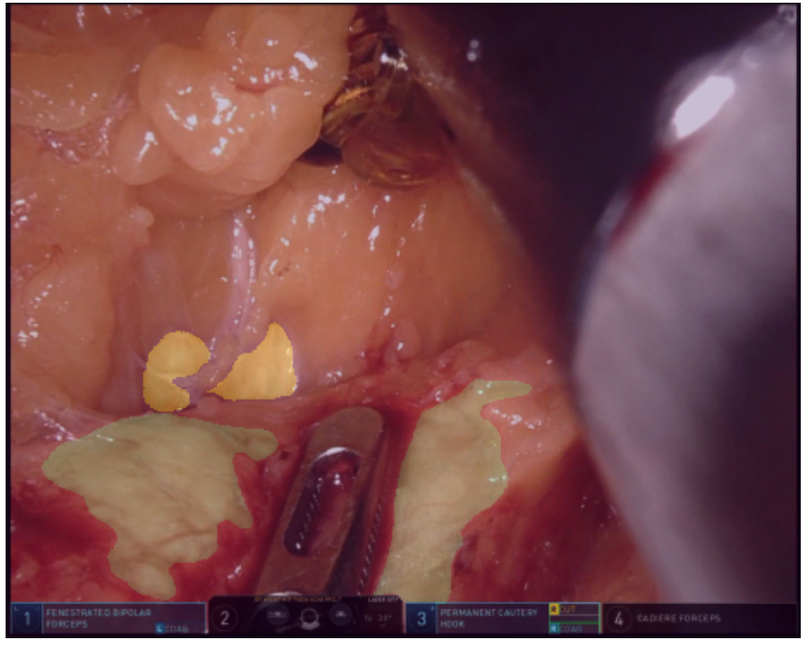
Reference Image



Ground Truth



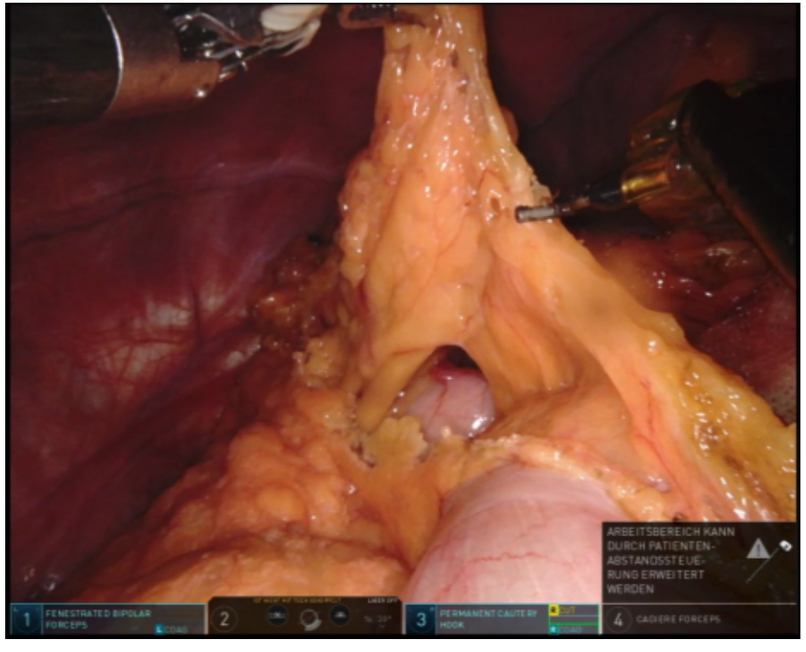
Ground Truth Overlay



Present classes: background, pancreas, stomach

4. Qualitative Analysis — train — Sample 563

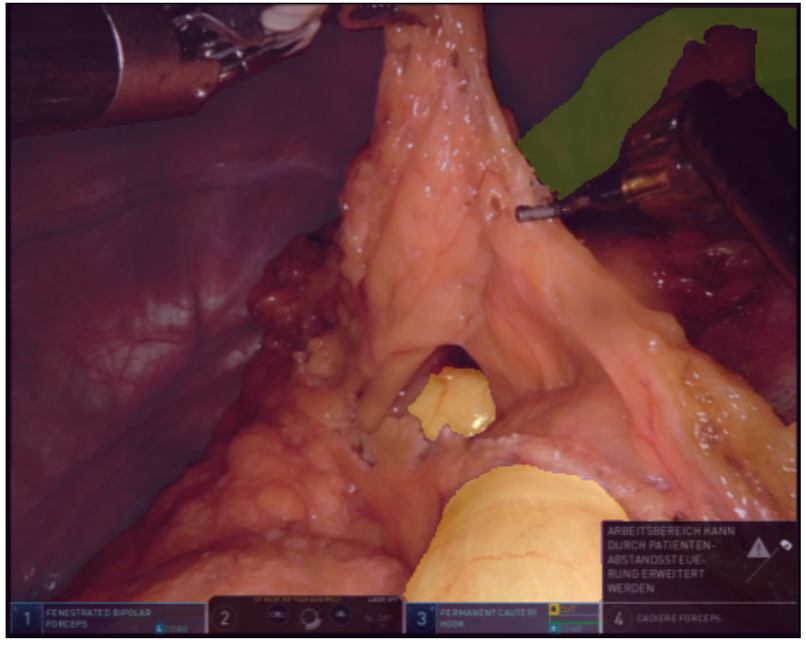
Reference Image



Ground Truth



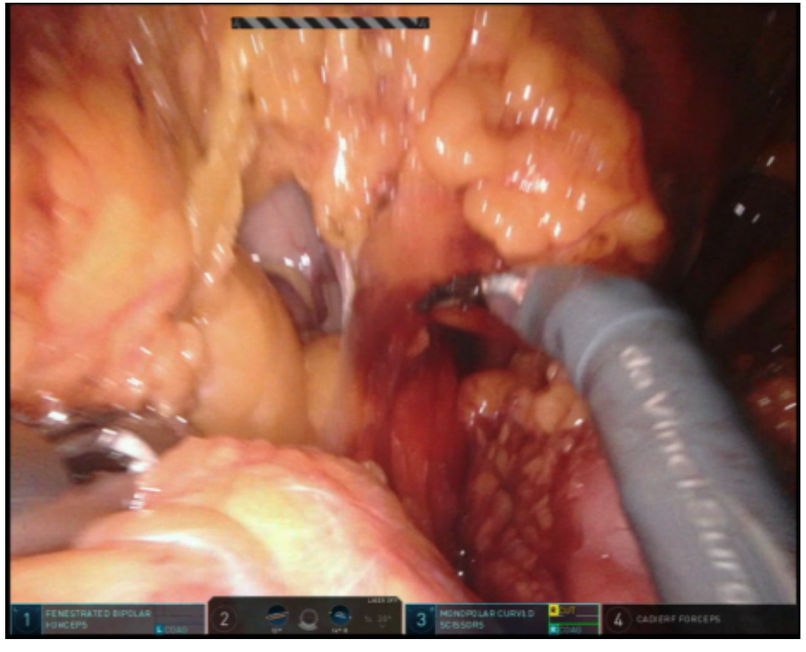
Ground Truth Overlay



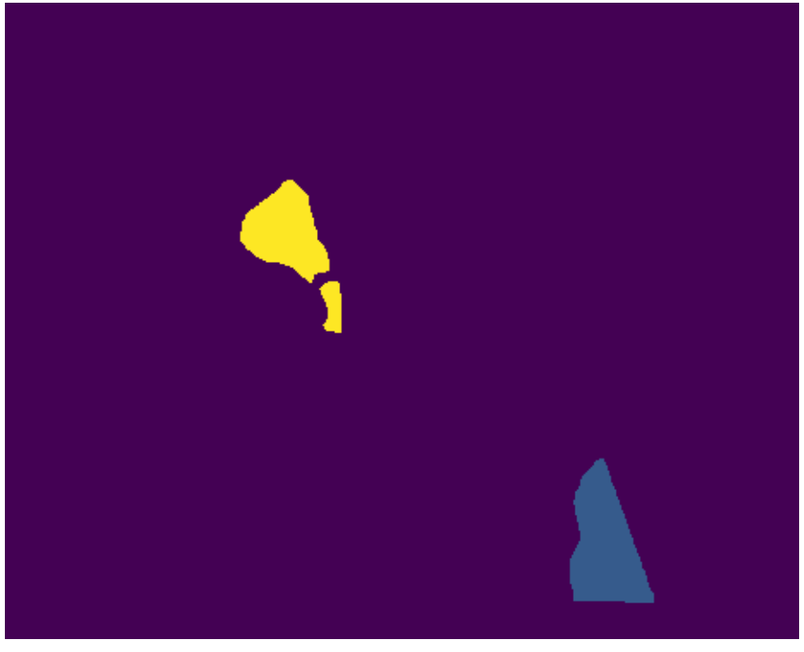
Present classes: background, abdominal_wall, spleen, stomach

4. Qualitative Analysis — train — Sample 76

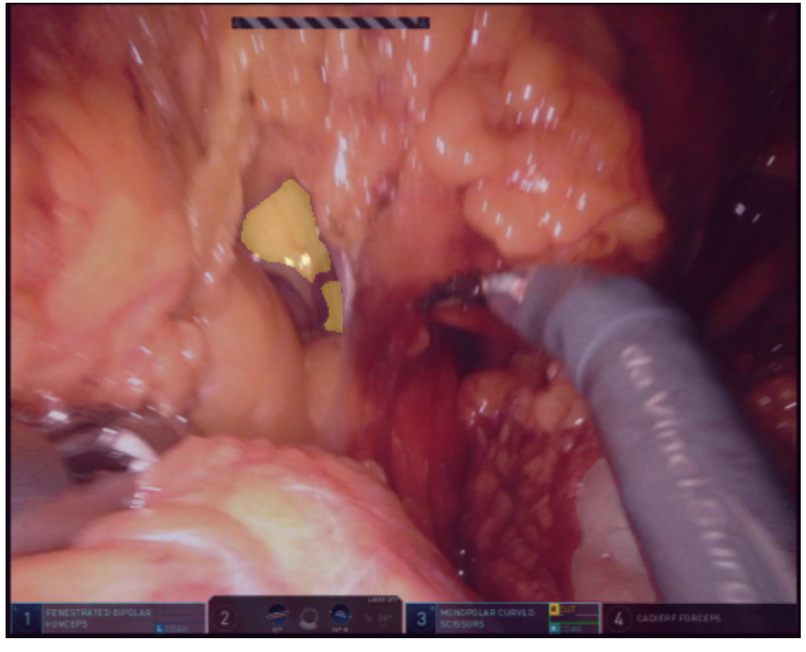
Reference Image



Ground Truth



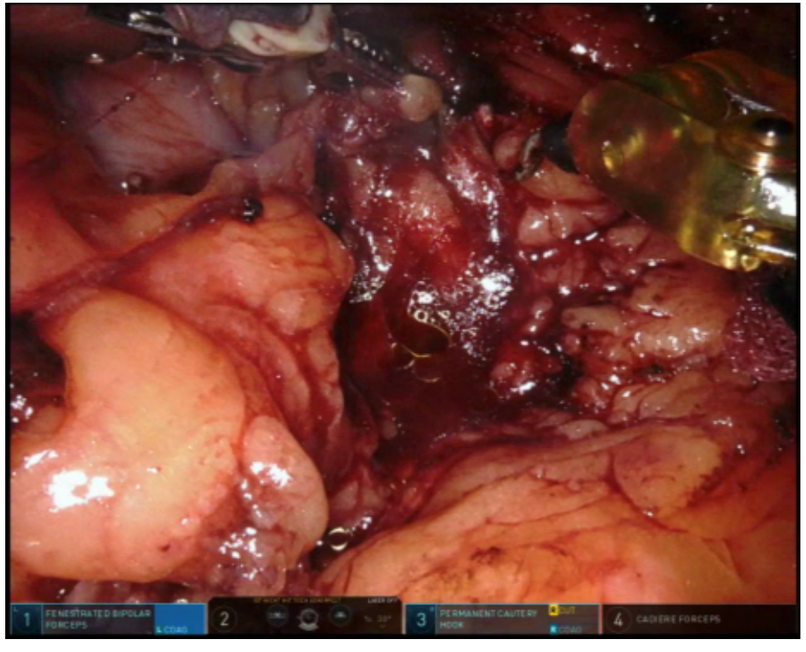
Ground Truth Overlay



Present classes: background, colon, stomach

4. Qualitative Analysis — train — Sample 373

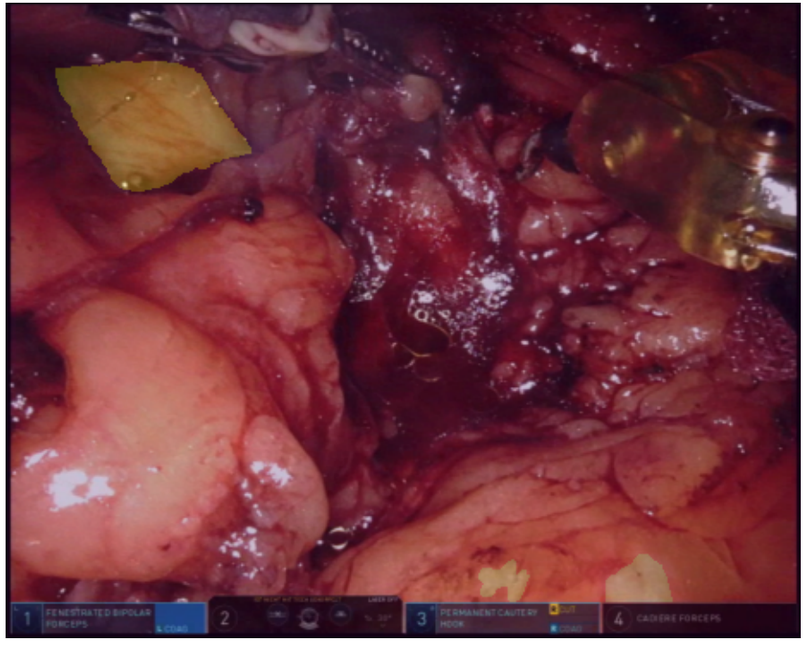
Reference Image



Ground Truth



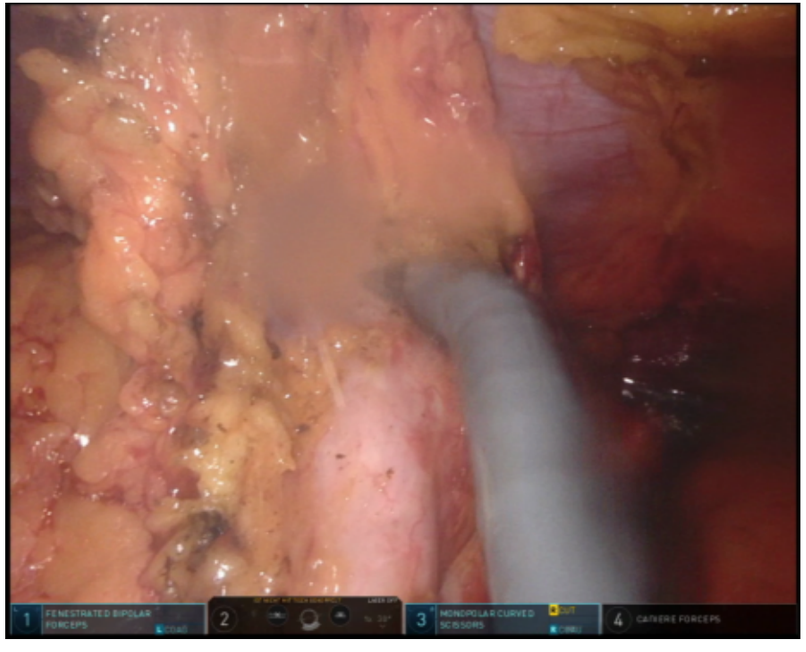
Ground Truth Overlay



Present classes: background, pancreas, stomach

4. Qualitative Analysis — validation — Sample 1017

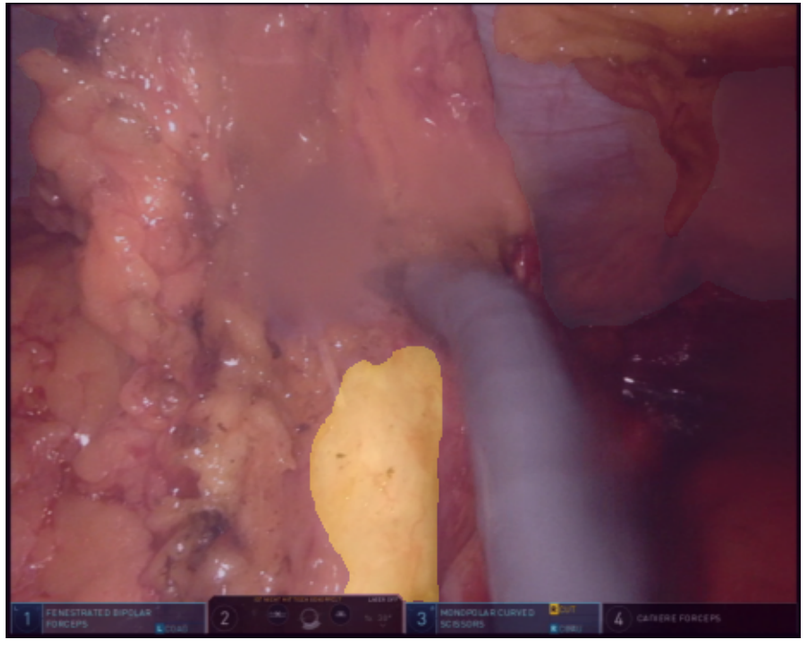
Reference Image



Ground Truth



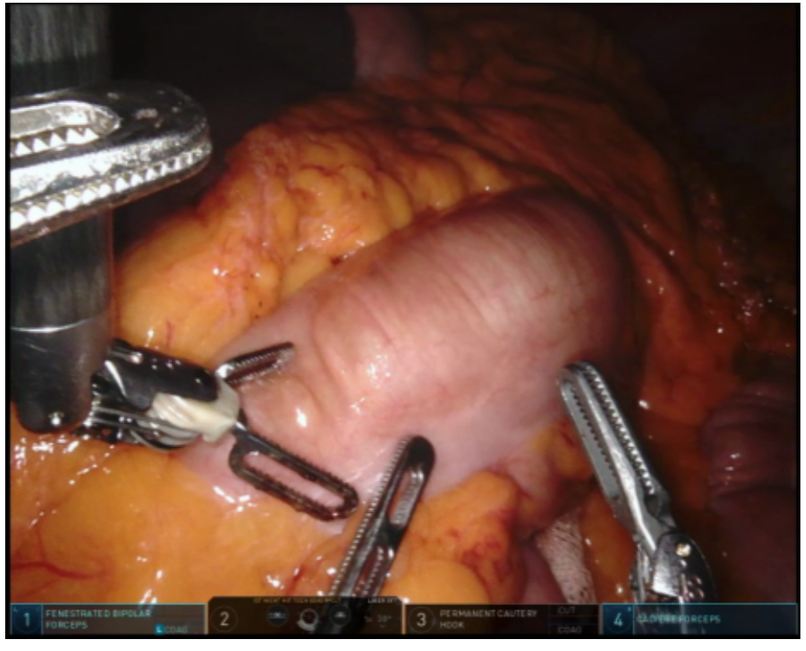
Ground Truth Overlay



Present classes: background, abdominal_wall, stomach

4. Qualitative Analysis — validation — Sample 951

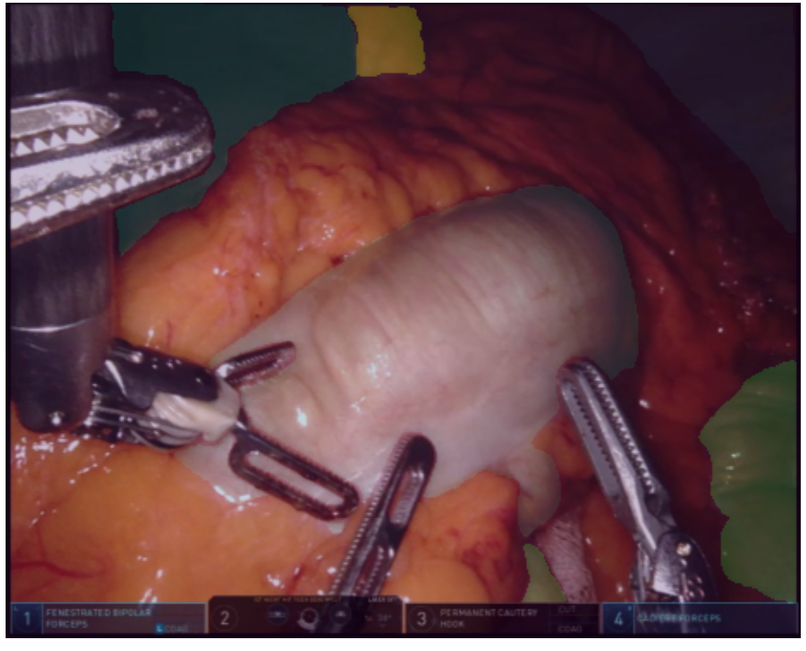
Reference Image



Ground Truth



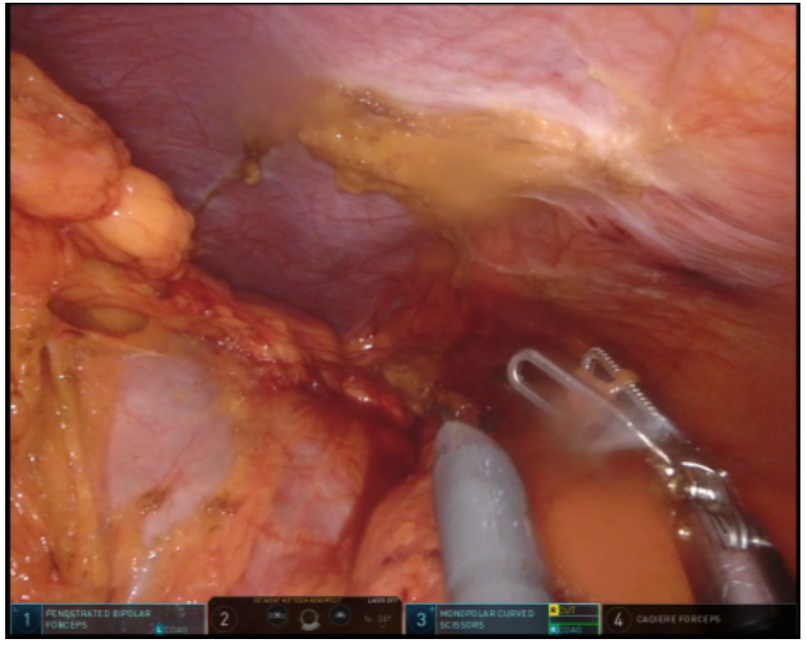
Ground Truth Overlay



Present classes: background, abdominal_wall, colon, liver, small_intestine, stomach

4. Qualitative Analysis — validation — Sample 993

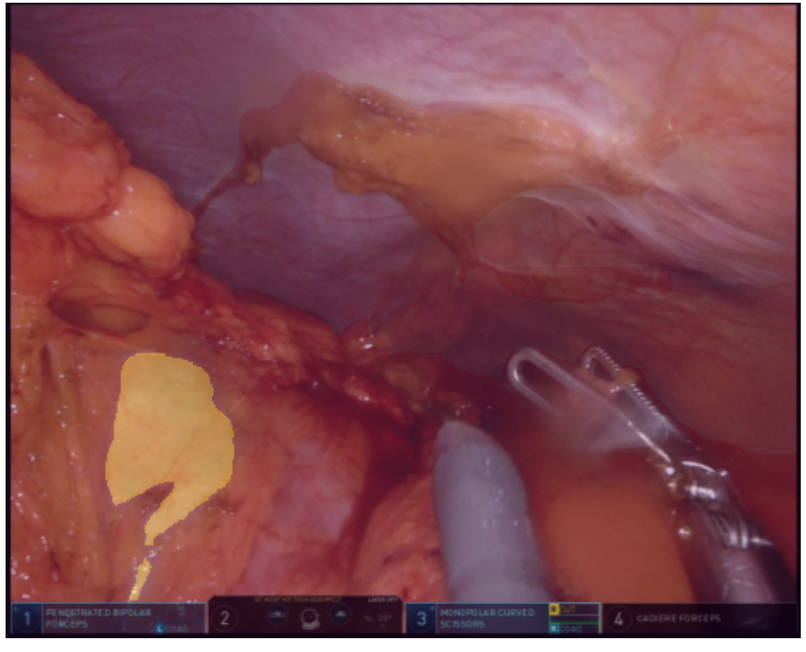
Reference Image



Ground Truth



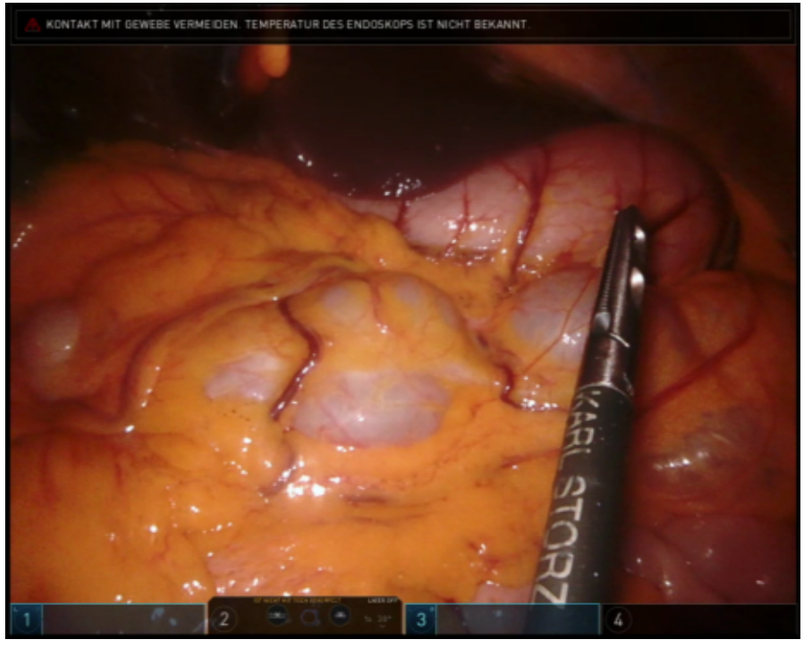
Ground Truth Overlay



Present classes: background, abdominal_wall, stomach

4. Qualitative Analysis — validation — Sample 880

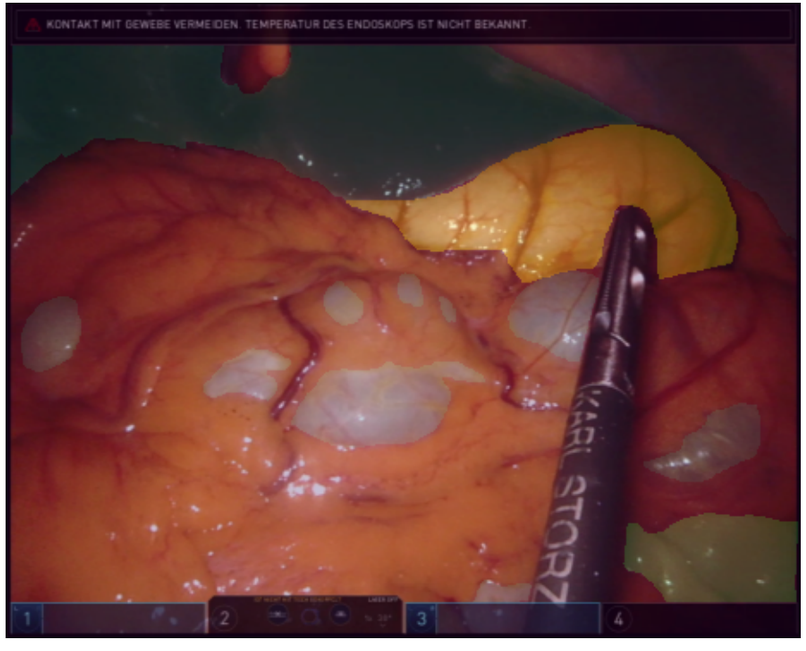
Reference Image



Ground Truth



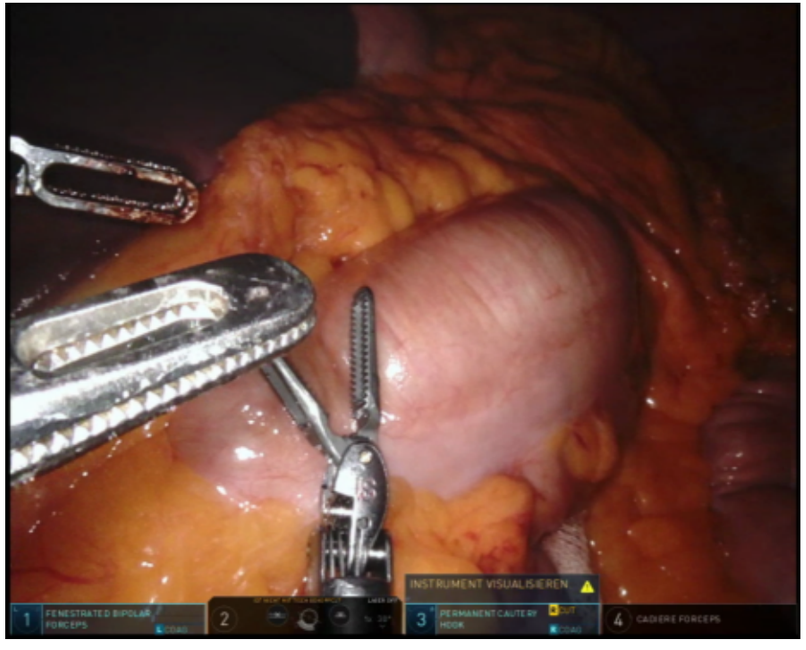
Ground Truth Overlay



Present classes: background, abdominal_wall, colon, liver, small_intestine, stomach

4. Qualitative Analysis — validation — Sample 950

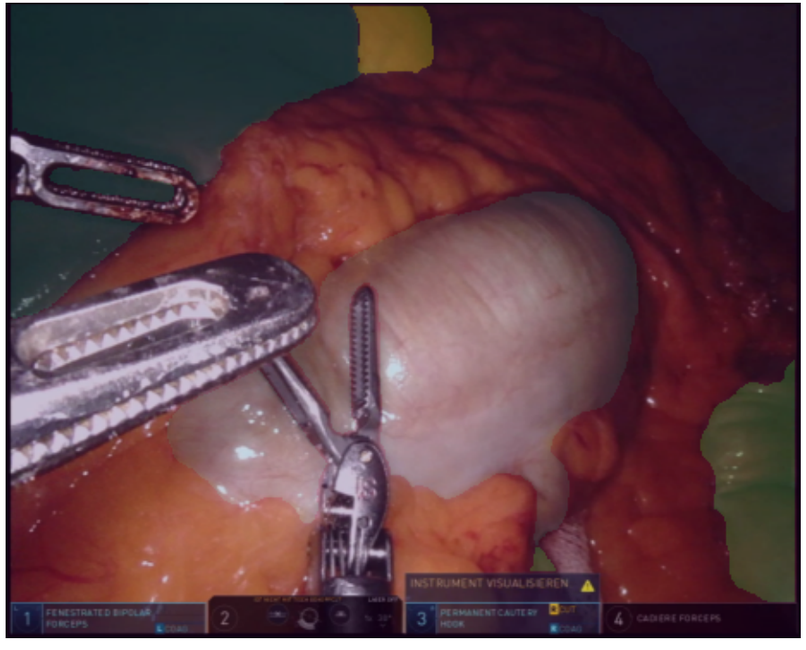
Reference Image



Ground Truth



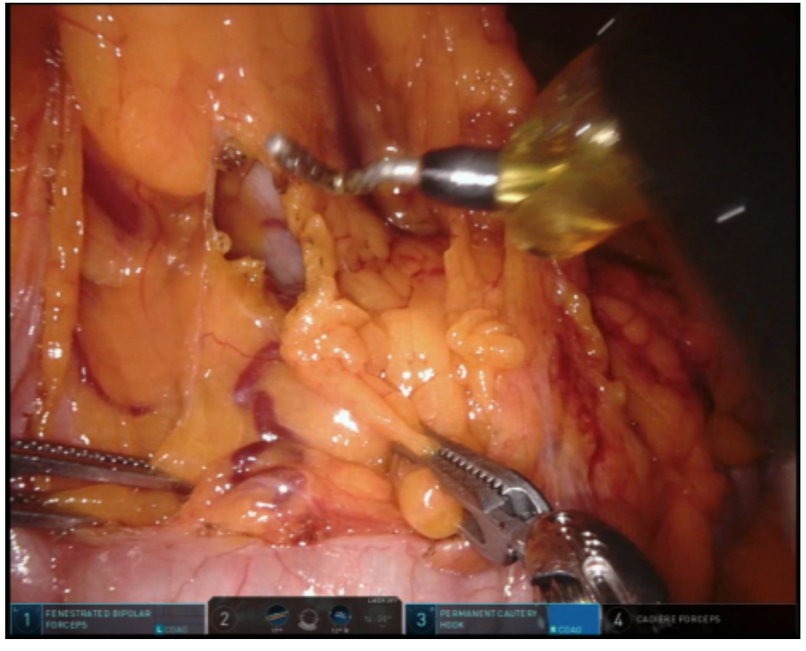
Ground Truth Overlay



Present classes: background, abdominal_wall, colon, liver, small_intestine, stomach

4. Qualitative Analysis — test — Sample 1345

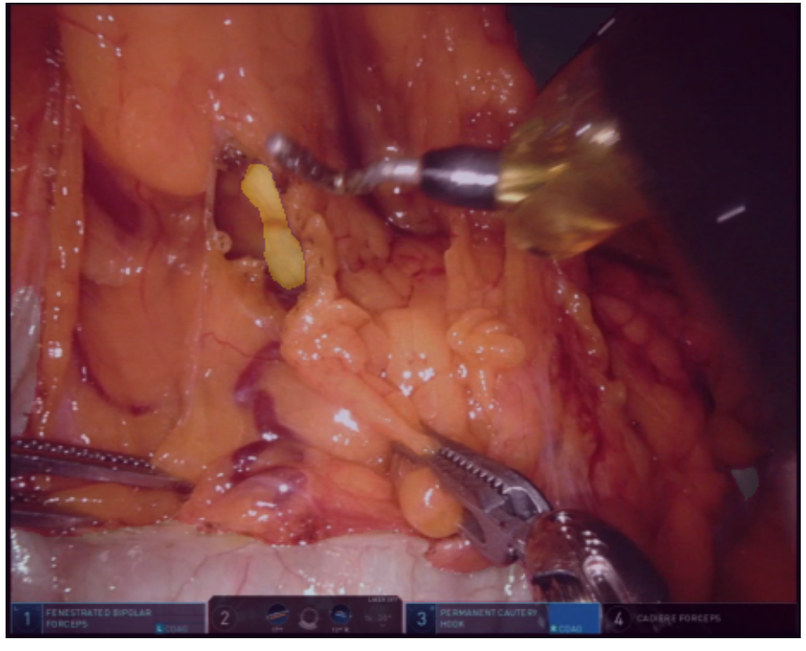
Reference Image



Ground Truth



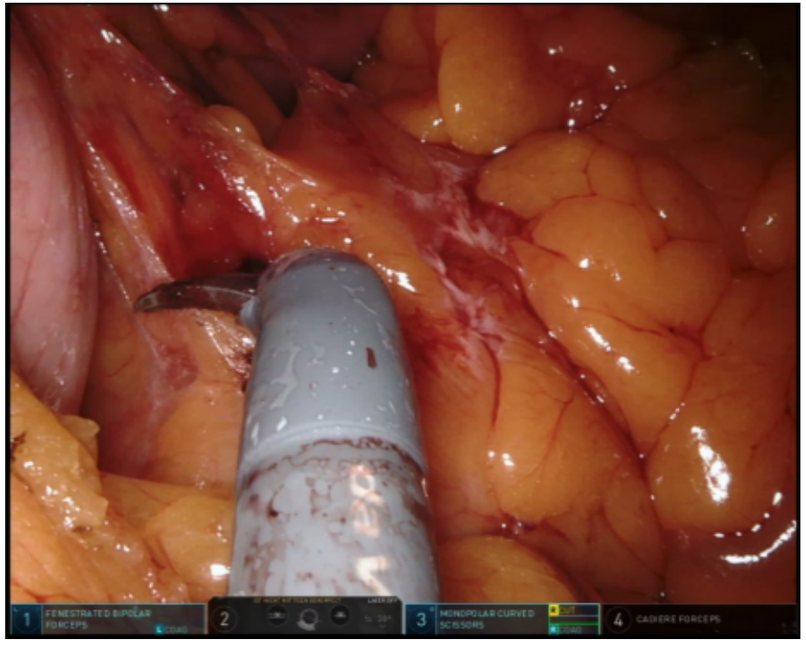
Ground Truth Overlay



Present classes: background, abdominal_wall, colon, stomach

4. Qualitative Analysis — test — Sample 1224

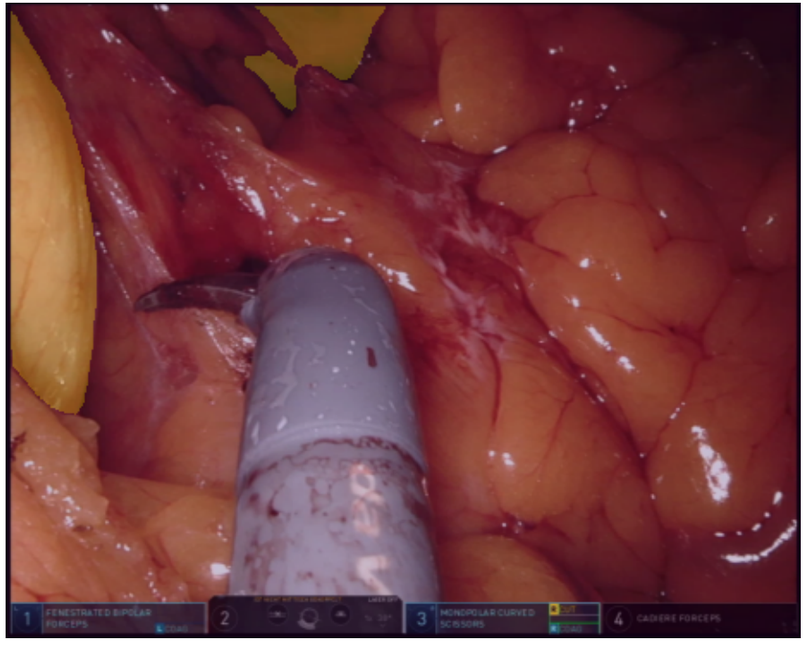
Reference Image



Ground Truth



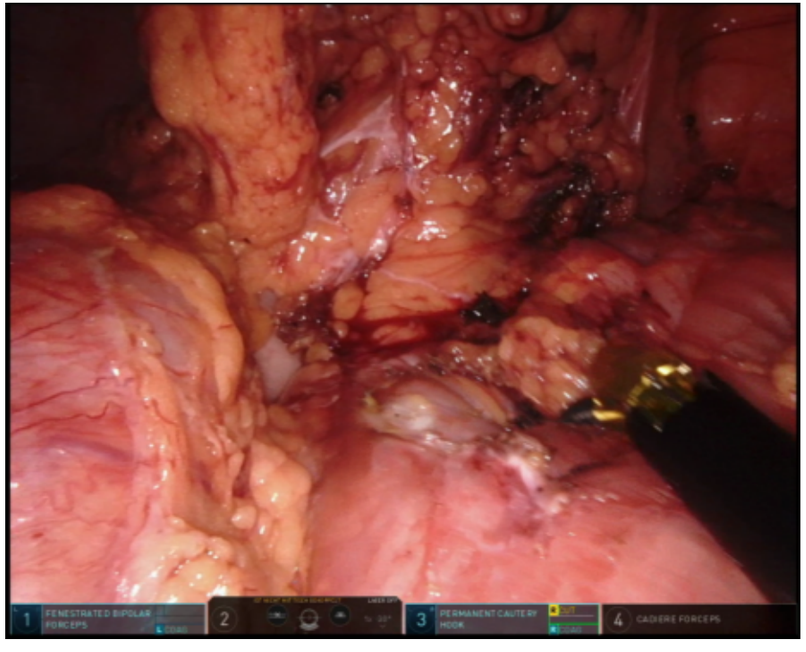
Ground Truth Overlay



Present classes: background, stomach

4. Qualitative Analysis — test — Sample 1302

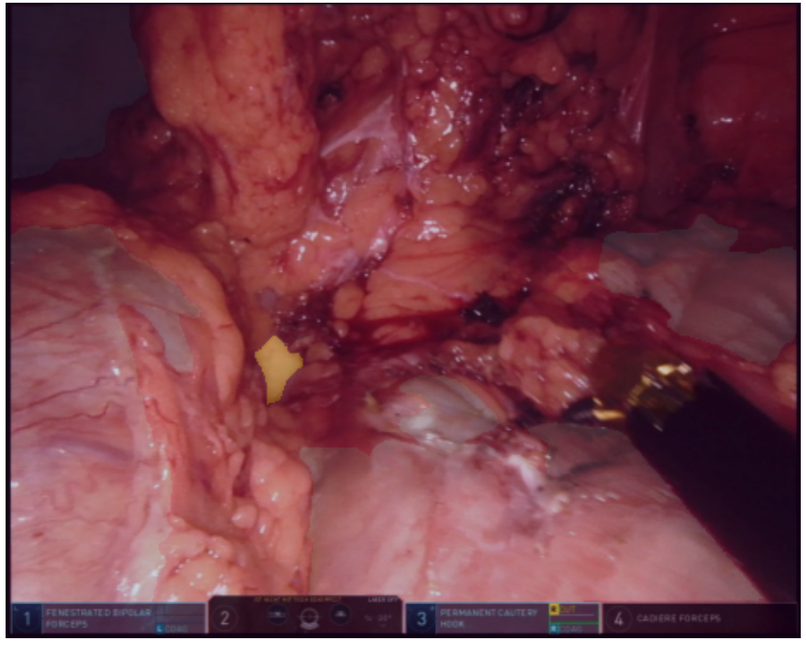
Reference Image



Ground Truth



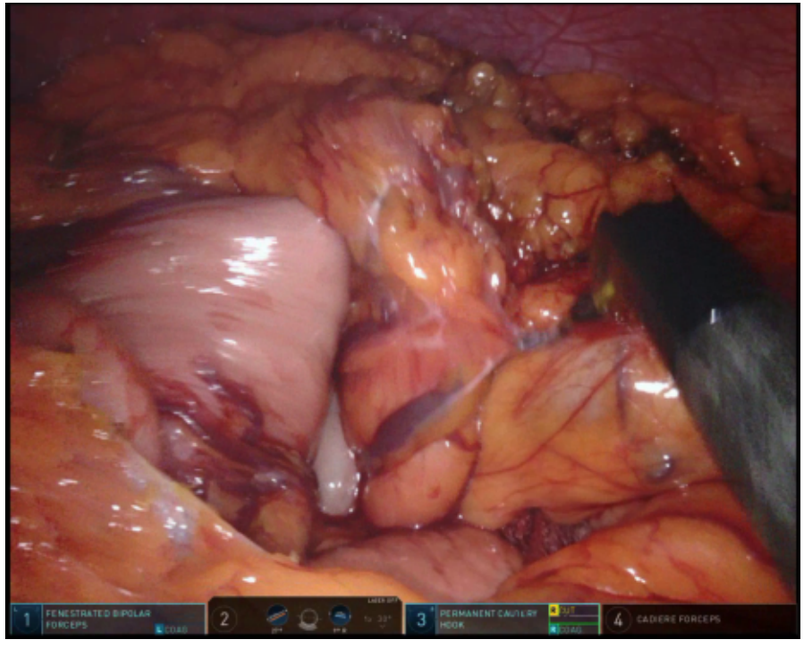
Ground Truth Overlay



Present classes: background, abdominal_wall, colon, stomach

4. Qualitative Analysis — test — Sample 1097

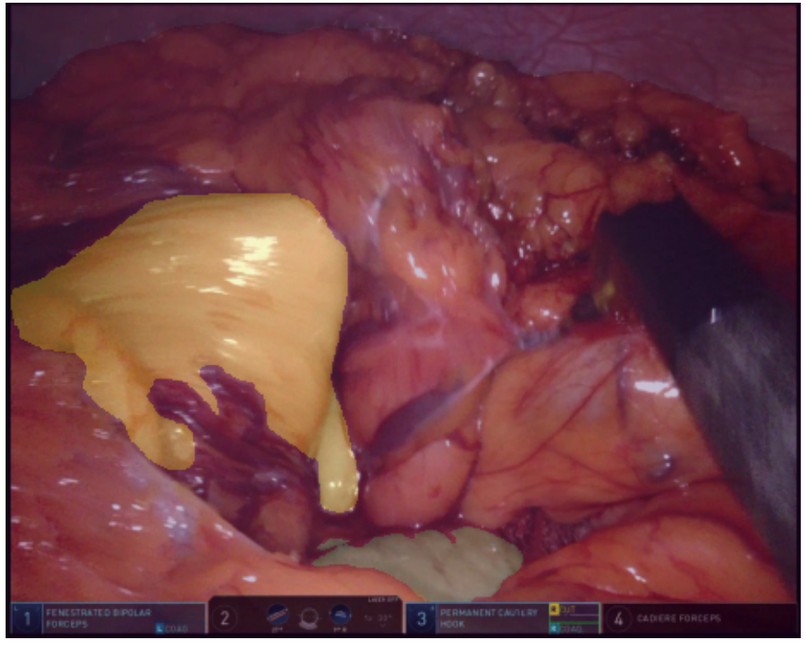
Reference Image



Ground Truth



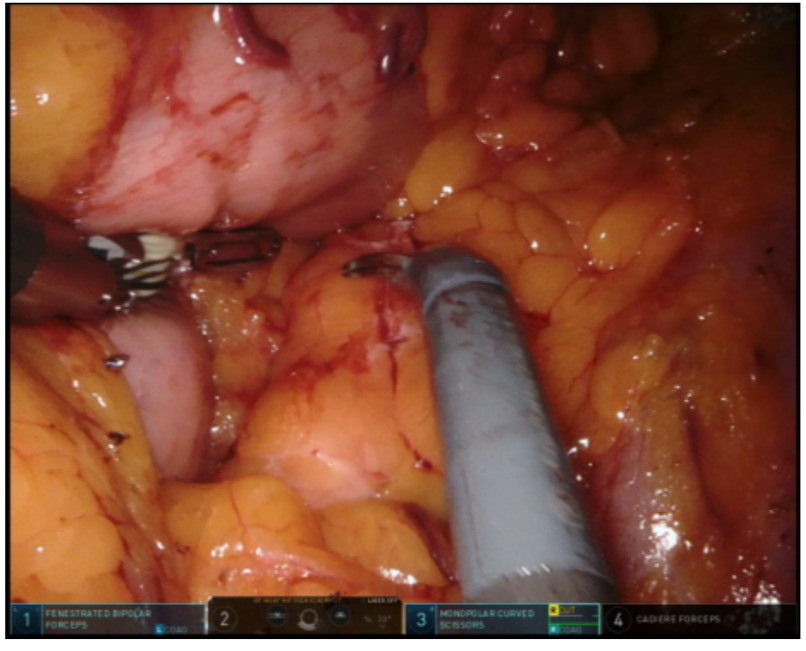
Ground Truth Overlay



Present classes: background, abdominal_wall, pancreas, stomach

4. Qualitative Analysis — test — Sample 1223

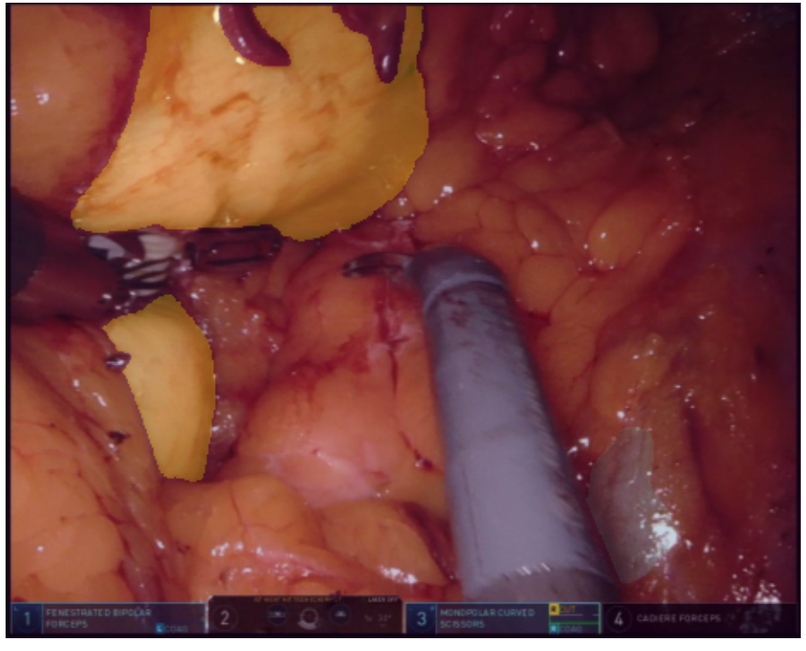
Reference Image



Ground Truth



Ground Truth Overlay



Present classes: background, colon, stomach

5. Validation Summary

Dataset: dsad

Number of classes: 8

Split sizes:

- train: 863 samples
- validation: 202 samples
- test: 365 samples

Class presence:

- train: small_intestine has the lowest sample presence (6.60%).
- validation: spleen has the lowest sample presence (0.00%).
- test: spleen has the lowest sample presence (0.00%).

Pixel distribution:

- train: background is the dominant class (81.62%), while pancreas has the smallest pixel share (0.2474%).
- validation: background is the dominant class (74.05%), while spleen has the smallest pixel share (0.0000%).
- test: background is the dominant class (83.14%), while spleen has the smallest pixel share (0.0000%).

Interpretation:

- Sample presence measures how frequently a class occurs across images.
- Pixel distribution measures how much image area is occupied by each class.
- These two statistics describe different forms of class imbalance.
- Differences between splits should be considered when interpreting model performance.
- Group-level stratification cannot guarantee that every rare class appears in every split.